

Write a Prolog Program to Implement Medical Diagnosis Expert System

Aim

To develop a simple Prolog expert system that diagnoses diseases based on the symptoms provided by the user.

Objective

- To understand expert systems in Artificial Intelligence.
- To represent medical knowledge using facts and rules.
- To infer diseases based on symptoms.
- To perform logical reasoning using Prolog.

Theory

An **Expert System** is an AI application that mimics the decision-making ability of a human expert. In Prolog, expert systems are built using **facts** and **rules**. Symptoms are represented as facts, while diseases are represented as rules. When a user enters symptoms, Prolog compares them with the knowledge base and suggests the possible disease.

Algorithm

Step 1: Define symptoms as facts.

Step 2: Define diseases using rules.

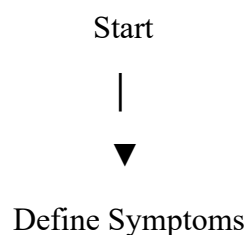
Step 3: Enter the patient's symptoms.

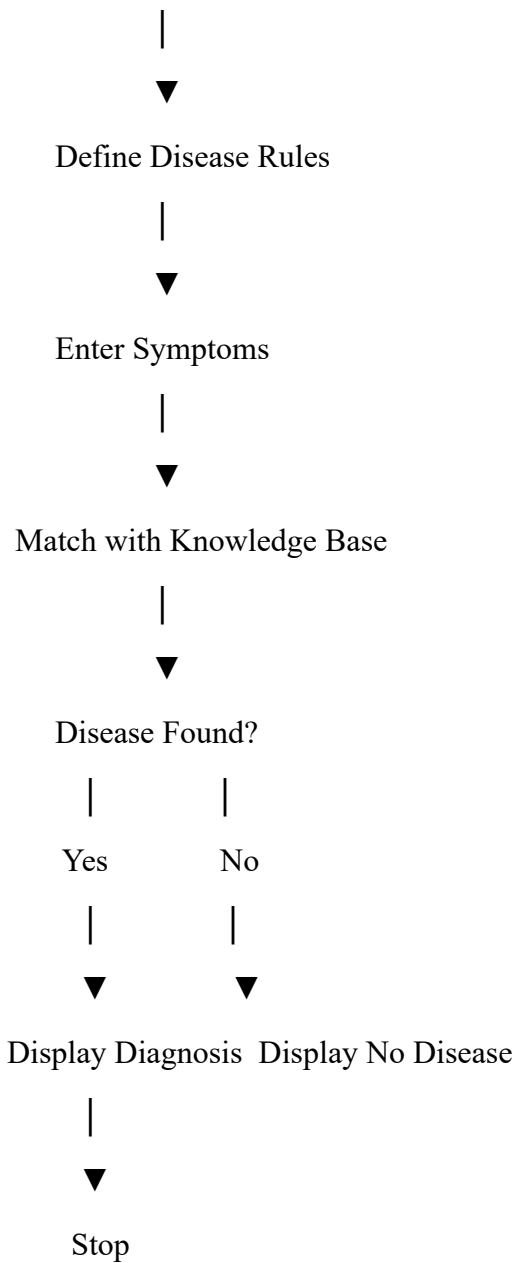
Step 4: Match the symptoms with the knowledge base.

Step 5: Infer the possible disease.

Step 6: Display the diagnosis.

Flowchart





Prolog Program

List of symptoms

```
symptoms = ["fever", "cough", "headache"]
```

Function to check diseases

```
def disease():
```

```
    diseases = []
```

```
# flu :- fever and cough
if "fever" in symptoms and "cough" in symptoms:
    diseases.append("flu")

# migraine :- headache
if "headache" in symptoms:
    diseases.append("migraine")

return diseases
```

```
# Display diseases
result = disease()

print("Possible diseases:")
for d in result:
    print(d)
```

Sample Output

?- disease(X).

X = flu ;

X = migraine.

Result

The Prolog program successfully identified diseases based on the symptoms stored in the knowledge base.

Conclusion

[illegible]

```
PS C:\Users\Varshini M\OneDrive\Desktop\Here!!!ARTIFICIAL INTELLIGENCE\VC EXP> & "C:\Users\Varshini M\AppData\Local\Programs\Python\Python314\pyt
PS C:\Users\Varshini M\OneDrive\Desktop\Here!!!ARTIFICIAL INTELLIGENCE\VC EXP> & "C:\Users\Varshini M\AppData\Local\Programs\Python\Python314\pyt
Possible diseases:
flu
migraine
PS C:\Users\Varshini M\OneDrive\Desktop\Here!!!ARTIFICIAL INTELLIGENCE\VC EXP> & "C:\Users\Varshini M\AppData\Local\Programs\Python\Python314\pyt
Possible diseases:
flu
migraine
PS C:\Users\Varshini M\OneDrive\Desktop\Here!!!ARTIFICIAL INTELLIGENCE\VC EXP> & "C:\Users\Varshini M\AppData\Local\Programs\Python\Python314\pyt
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flu
migraine
```